

Performance Optimization in Cloud Computing using Machine Learning

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Abstract-

This research shows how time plays an important role when it comes to smart resource use, scheduling and allocation in cloud computing. This study differentiates between two machine learning models: Support vector machine (SVM) and Decision tree (DT), for estimating the runtime based on CPU usage, memory use, disk speed, and network delay using MATLAB. SVM tries to predict the smallest possible error between real life and predicted outcome, whereas the DT (Decision tree) groups execution time into groups or bins (0-10 sec, 10-20 sec, etc.), and they are being used to predict time. To compare which model is better there are two standard tests for that- Firstly there is this MEAN SQUARED ERROR (MSE) which tells the how accurate are your predictions. Secondly, we have the r- squared method that tells how well the model demonstrates the variation in data. Results shows that the SVM is far better than DT by a stunning accuracy of 81%. Thus, SVM got much better predictions, that are pretty close to the actual values, while the DT had predictions which likely to have more error than that of SVM. And it is believed that in upcoming years there might be a combination of SVM with another cloud models to get even better results and predictions.

Keywords- *Cloud computing; Machine learning; Support Vector Machine; Decision Tree; Prediction; Runtime; Network delay; Mean squared error.*

I.INTRODUCTION

Cloud computing is a game changer in the modern IT, it allows the user to access storage that is not currently in the device storage but somewhere in the cloud that can only be accessed online. It is like a service that is available anytime in the internet. And with these advantages Cloud services are gaining more and more popularity due to efficient memory allocation and deduced workload execution time due to which minimized costs and maximized performance by the cloud providers can

be seen. The workload execution time is defined as the time taken to complete any task in the cloud. But this is affected on several factors like- CPU utilization, workload distribution, speed of reading and writing data and latency as well. If these all things are done in order, then there will be a significant boost in the efficiency which saves cloud resources and make their usage in smart and balanced way.

As necessity is the mother of invention, therefore with the exponential growth of cloud computing, cloud servers are made to handle huge amount of data and are able to process it. Clouds are used to handle loading of data and allocating it, that is used by huge tech companies. To make these allocations more efficient we need newer models as older scheduling methods like the elastic computing usually don't adjust well in real time resource allocation. So, machine learning can provide a remedy for it, as it can use past data of users which provides a better resource planning than previous. Now there comes two widely used ML models for this prediction which are SVM and DT. They are popular because they are easy to understand and handle data effectively. SVM is good at predicting outcomes using math-based models, whereas the decision tree works as flowcharts that is easy to interpret. Still there is a dilemma of choosing which ML model suits best for predicting a cloud task.

Many researchers have studied how ML can be used to manage cloud resources, but not all problems are sorted till now, which include-

- (i) Most of the studies use only one ML model at a time, which limits the comparison between other models, thus we are not able to find which works best for the prediction of how long tasks take.

- (ii) Many current ML models need fine tuning and a large dataset, meaning they should be more accurate and efficient.
- (iii) Some researchers are based on very particular type of cloud system that is not portable i.e. the results cannot be displayed everywhere. But a promising model should work in all conditions.
- (iv) Many models don't really care about factors like-CPU, memory, latency, and disk read and write speed. And is they are being ignored then this may cause less accurate predictions.

This research aims to fill the gap mentioned earlier with following aims:

- (i) To evaluate both the ML models, i.e SVM and DT, for predicting how long will a task take in the cloud to run and complete.
- (ii) To check out of the twos which model performs better and gives better results by using standard metrics like MSE and R^2 .

It will thus check how important is different cloud resources like CPU, memory, etc. And shows how they affect prediction accuracy. This study will also explain the positive and negative part of both of the models and how they affect cloud resource planning.

With cloud computing growing rapidly, it's become more important than ever to manage resources smartly. This project dives into how machine learning can help predict how long tasks take to run in cloud systems. By comparing two models—Support Vector Machines (SVM) and Decision Trees (DT)—we aim to find which one gives better and more accurate predictions. These predictions can help cloud providers reduce delays and assign resources more efficiently. This study also checks how some factors like CPU usage, memory, disk speed, and network delay can be the reasons that affect the accuracy of these models. Lastly, this research provides a clearer picture of the pros and cons of each model and sparks the idea of using hybrid ML methods in the future to make cloud performance even better than previous.

II. LITERATURE SURVEY

Back in the days cloud systems use Elastic computing, that technically rents or provide a specific ratio of cpu, ram and storage, so that all the work can be stored in the internet, rather

than filling up the devices. But this leads to poor resource management as we have the fix ratio of cpu and RAM, and sometimes requirements won't meet according to the app we are working on. To counter this, we have Flexible computing that allocates resources based upon the app's need, and also solve noisy neighbor effect. It is seen that FC uses 33% less resources than EC [1].

Cloud computing is at its peak at the moment, and are required in different IT sectors. But which is perfectly match the organization is a big question, as there are many cloud service platforms like AWS, Microsoft Azure, GCP, and many more. Researches came up with the conclusion that all the platforms are best in their respected fields, like Amazon Web service (AWS) can handle heavy stuff, Microsoft azure is for enterprise integration, google cloud platform that fits for machine learning. So rather having a debate over the platforms one should it according to the needs [2].

Cloud and hybrid cloud are the top priority for an e commerce companies, it doesn't only strengthen them but also provides them flexibility. For which an integration of AI (artificial intelligence) and GI (general intelligence) is to be used to enhance cloud and hybrid computing by providing maximum energy at the time of maximum use, and reducing it at the time of less user activity [3].

There can be a new system or an algorithm for processing large amount of data which can be executed by using data partitioning, parallel computing and uniform distribution of work load. Parallel computing is designed so that multiple tasks can run simultaneously, that can reduce the waiting time. Distribution of work or compression can also increase the efficiency. It is seen that after compression of 1TB data, at 85% then the processing efficiency increases by 60%. This can b applied not only in engineering chiefly, but also in finance, logistics and healthcare also [4].

There are various other models of cloud computing as well like "fog" and "edge computing" which are pay-as-you-go models, resembling the prepaid mobile plan we use i.e, the company or the organization will pay only for the time and resources they have used. This reduced unnecessary wastage of commodities [5].

Researchers also suggest that when hybrid (public+private) computing and traditional computing are mixed together, they

can be moulded to form into a next generation system, that are designed to solve complex problems faster and efficiently. But sadly, it cannot be built due to compatibility issues, but if a hybrid computing system is made it can revolutionize many tech companies for sure [6].

With a greater number of devices, there comes the problem of delaying tasks. When a task runs, an executor(intermediator) is created. But building these executors is time taking. So, to overcome this, the author built a system driven by A.I called “Reflex Pilot”, that processes faster by 71% [7].

Furthermore, cloud computing can be used in handling geospatial data, that deals with government, urban planning and disaster management. But since cloud is far away from users and therefore a latency can be observed, but the information it is sharing must be fast and with no latency. To sort this thing out MEC (Multiaccess Edge Computing) came into picture, that reduces latency since MEC servers are placed close to the user [8].

Cloud also helps in the field of healthcare, by managing patient’s data and stats. But these can be prone to leaks and all, for which blockchain access control come into picture that only lets the verified users access their information. Stats show that this method blocked 99.97% of the DoS attacks and 99.99% integrity in the system, by keeping the HIPAA rules intact [9].

This low latency problem arises in IoV (internet of vehicles) as well. For which we can integrate A.I into our cloud as a mini computer to minimize delay and to show that this method works faster than the old school methods [10].

This research shows how cloud computing helps in the improvement in the management of renewable energy system, like water, sun, wind and many more. Its core work is to improve energy efficiency, formatting and workload distribution, that significantly increase the efficiency level [11].

Cloud computing can also resolve the issues of resources allotment, for example if there are many requests in a site, then it is hard to assign each and every of them with what they want. To counter this there exists an algorithm named “Resource Aware scheduling”, and with the availability of resources, priority wise, and work load it will allot the required resource [12].

This paper discusses about a framework to tackle task scheduling in edge computing system. It includes- TO-PS algorithm that can run parallel computing and can do heavy tasks easily. And for the security terms we have “PE-ACS” to protect data from cyber attack. All these compiled up can increase performance and reduce energy and a 96.7% success rate in protection of data [13].

Brushless DC motors are extensively popular as they are highly dependable and require less maintenance cost. To make the best out of it it’s important to control their speed accurately when connected to a variable load output. This project comprises of both neural network and pi controller which is built in MATLAB for improving performance of BLDC motor. The behavior exhibited by motor is determined by using maths equations that can relate emf, current, torque and voltage. Tests were simulated to check how well the motor run in different scenarios. And showing its major improvement in important areas like stability, latency, etc. [14]

This work basically presents a fuzzy logic based on Dynamic Spectrum Allocation (DSA) system developed in MATLAB to handle various interference, traffic, and user needs in wireless communication. A Fuzzy Inference System (FIS) has been designed with rules to adaptively allocate spectrum. The system is validated using performance metrics like MAE, MSE, RMSE, and R^2 , showing high and accurate predictive accuracy. The results demonstrate the stiffness and clarity of fuzzy logic in efficient spectrum management [15].

This study is roaming around two ML models, KNN and PNN in MATLAB in order to classify person’s heart in three groups: risky moderate and good. The classification is based on person’s heart rate and his cholesterol level. Since heart is the most vital organ in the body it is good to keep a track of it, this study aims to accelerate this idea, by creating a computational method that is accurate and fast, along with their pros and cons in the medical field. PNN can handle noisy data but requires more computing power thus less flexible whereas KNN is simple but it’s slow when predicting and can be affected by noisy data [16].

This study improves the accuracy and performance of text classification. KNN due to its simplicity is widely used for text classification. In this approach each document is converted into a vector based on some selected keywords or features. In the traditional KNN, all words are treated equally that doesn’t seem

to categorize them. To fix this, this study came up and boosts KNN so that it can adjust keywords according to their categories [17].

This article put forward a smart method using an Associated Recurrent Neural Network (ARNN) to detect which devices in an IOT is compromised or hacked by analyzation. The ARNN learns pattern of normal and abnormal behavior using Artificial Bee Colony Optimization for feature selection that does not require any separate detectors for each device. It's tested on the real-world data and can outperform existing techniques [18].

III. RESEARCH METHODOLOGY

Machine learning methods are being used in this study to determine how long a task takes to complete in cloud environments. This methodology consists of Creating Data, Training machine learning models, checking their performance and plotting their results either in graphs or tables.

Data Processing:

A secondary data of 100 samples is used here to see how workload behaves in the cloud including parameters such as CPU usage, memory utilization, network latency and disk read and write speed. 80 samples were made to train the model and 20 were used to see how well they can perform on new unseen data.

Machine learning models:

For predicting the execution time more accurately, two machine learning models- SVM (Support vector machine) and DT (Decision Tree) were applied using MATLAB. Both of them were used to predict how long a task will take.

Support Vector Machine (SVM):

It is a powerful technique that establishes a connection between the input and execution time so that a hyperplane is identified so that errors can be minimized as illustrated in flowchart in Fig.1. Rather using tangled SVMs, simpler methods of minimizing errors were used. The equation stated above provides smooth output values by significantly reducing the error factor. SVM provides fine predictions, which is beneficial factor when workloads changes. It works the best on new data as it can find best match line rather than making stepwise

guesses. Linear connection between inputs and outputs results in better accuracy in SVM.

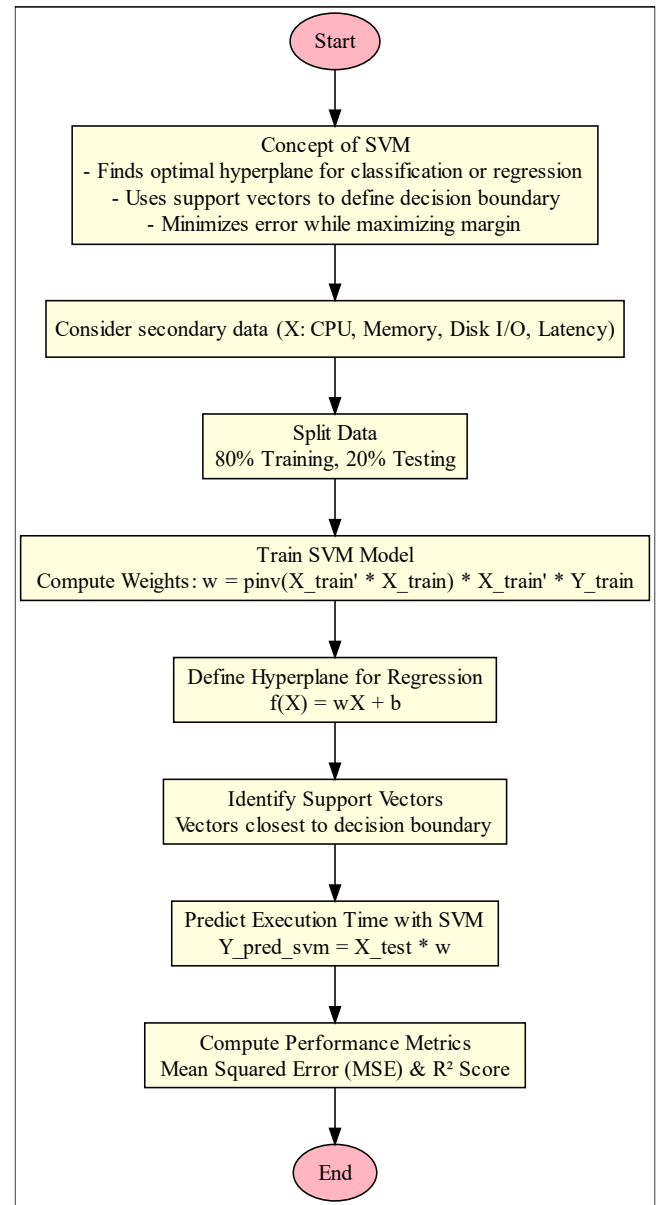


Fig.1 Flowchart for SVM model

Decision Tree (DT):

A basic model of decision tree was made by stacking times into 10 fixed ranges. And groups were divided on the basis of their average time result is shown. One by one predictions are made by DT, which makes it simple to understand. DT works well for yes/no type of data, and these DT results can be made better

using methods like cutting extra branches or combining trees as illustrated in flowchart in Fig.2.

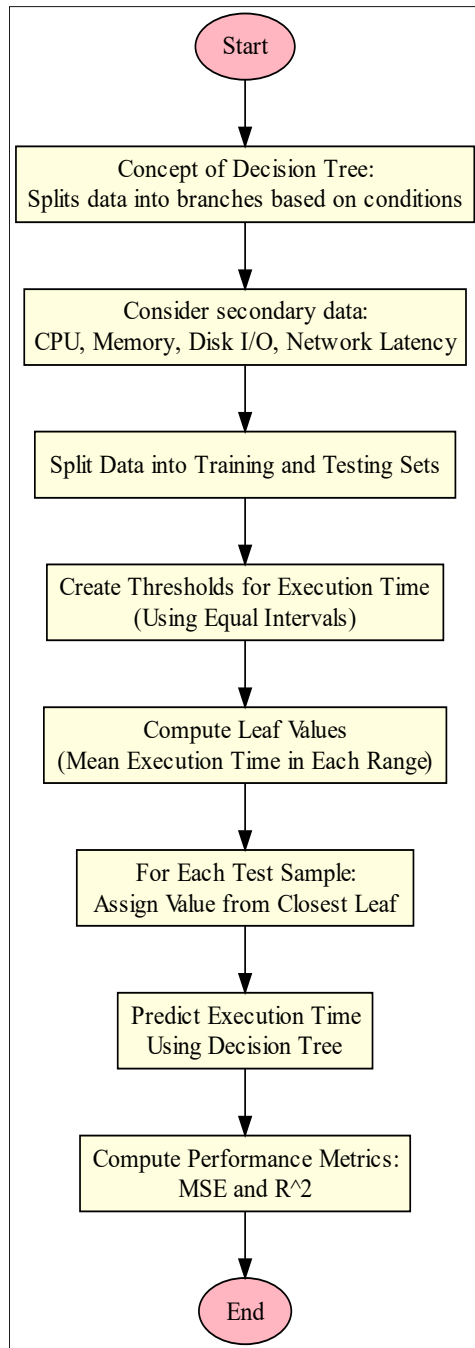


Fig. 2 Flowchart for Decision Tree model

Model Performance Evaluation:

The accuracy of both of the models were examined by using two known ways (SVM and DT). MSE (Mean Squared Error) checked how exact are the predictions away from the real values, and r-squared tell the variation of time explained by model- if it is closer to 1 this identifies the model to be better.

IV. RESULTS AND DISCUSSION

This particular section gives a brief difference between SVM (Support vector machine) and DT (Decision tree) models to evaluate how long do long tasks take to complete in cloud computing. And the results are checked using MSE (Mean squared error) and R-Squared score.

The TABLE I shows the accuracy score of both models. SVM got an MSE of 28.981 and R^2 of 0.98, whereas DT has an MSE of 81 and R^2 of 0.95, thus making SVM better prediction models than DT to predict execution time in the cloud.

TABLE I. PERFORMANCE COMPARISON

Model	MSE	R2_Score
"SVM"	28.981	0.98362
"Decision Tree"	81.004	0.95423

Fig.3 shows that how close are the predicted and real execution time is. As stated previously SVM is of more accuracy than that of DT model, precisely for some data points. This is a live proof that the SVM gives results that are reliable and of more precision when it comes to predicting the execution time in cloud. The R^2 score graph in Fig.4 also demonstrates the same, SVM is indeed more accurate. SVM's R^2 comes out to be 0.95 that states it can explain about 98% of the changes in data. Though R^2 for DT comes out to be 0.95 that is not bad, but is still less than SVM's accuracy. The MSE graph in Fig.5 shows that DT has more possibilities of error (81%) than that of SVM (28.981). This states that that DT is more prone to mistakes, most likely because it predicts using step-like methods that don't fit in the data as well.

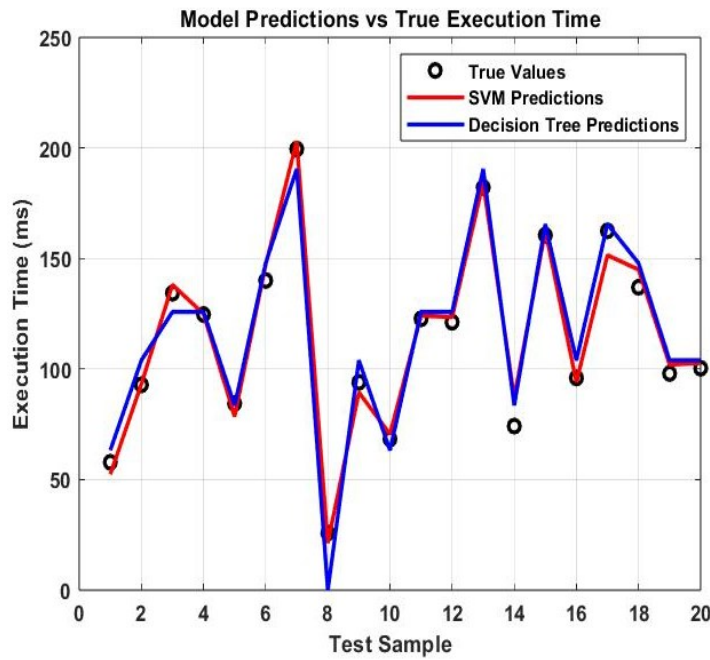


Fig.3 Comparison of predicted values

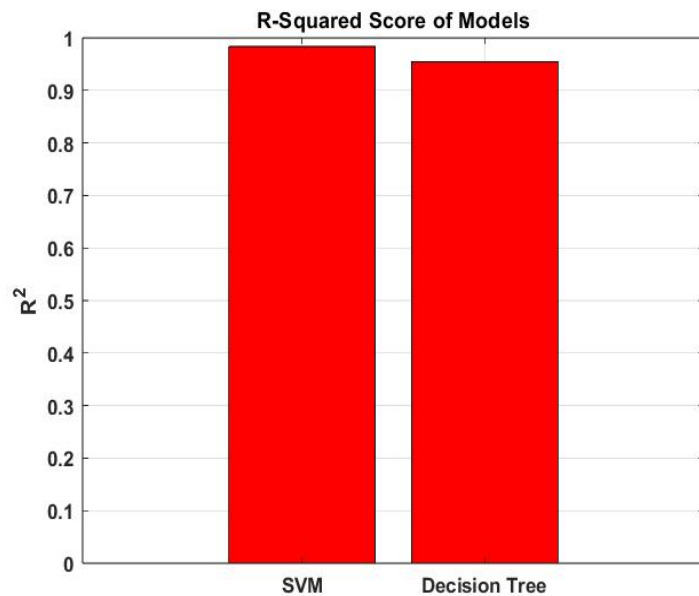


Fig.4 Comparison of R-Squared Scores

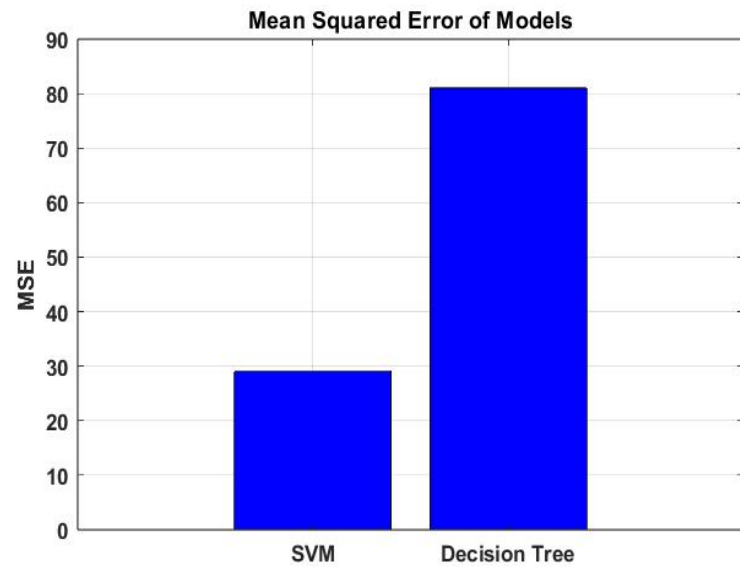


Fig.5 Comparison of mean squared errors

Key findings:

- (i) SVM provides better results than DT in computing or predicting execution time, with more accuracy and less errors.
- (ii) DT is not far away from SVM, which means it can be pushed forward with improved settings like better splits and pruning.
- (iii) SVM is a smooth prediction provider, which is better when we try to allot cloud resources correctly.

In the future we can test or implement different hybrid models that comprises of both conventional and non-conventional cloud systems so that accuracy of resource allotment may improve.

This test clearly lays down the fact that SVM is a better option to choose for predicting execution time in cloud computing. It gives better accuracy ($R^2=0.98$, $MSE=28.981$). But leaving DT behind shall not be a wise choice as it can be improved. In future projects, we can use deep learning or develop new algorithms in ML so that the results can be near perfect.

V. CONCLUSION

This research tells the ability of machine learning that can predict the time complexity of long tasks which takes place in cloud computing by using (SVM) support vector and (DT) decision tree models. A set of secondary datasets (around 100) were utilized with four important features like CPU usage, Disk read and write speed, memory use and network delay. Thus, the key results are DT was not bad than SVM, but SVM itself proves to be number one on the list by giving small error (lower MSE) and higher accuracy (higher R^2 score). SVM has smooth and continuous predictions, which makes it good for real time resource use. DT used step-based prediction, which is the reason for more errors and didn't display exact times. These all results are live proof that the SVM is way stronger in predicting cloud workload times, which helps in allocating and using resources wisely. And as stated earlier they might be a fusion of a hybrid cloud system that gives even better results and can revolutionize cloud systems.

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